Causal Propagation and Gravitational Waves from Projective Spectral Dynamics

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Abstract

We construct the Lorentzian completion of the spectral entropy framework in which the renormalized variation of a Laplace-type operator generates an infrared Einstein response. While the original formulation was elliptic and sufficient to derive the effective field equations and their thermodynamic interpretation, it did not address causal propagation. We define a Lorentzian spectral action via the real part of the hyperbolic determinant, equivalently through a Schwinger–Keldysh prescription, ensuring a real and retarded quadratic kernel.

Linearization about Minkowski spacetime shows that, in the regime $R\ell_\chi^2 \ll 1$ and $\omega \ll \ell_\chi^{-1}$, the theory propagates exactly two transverse-traceless tensor modes of helicity ± 2. Scalar and vector sectors remain non-dynamical in the infrared, and the additional pole generated by higher-derivative corrections lies at $k^2 \sim \ell_\chi^{-2}$, beyond current observational reach. The polarization content therefore coincides with that of general relativity in the accessible domain.

Beyond the leading Einstein sector, the ratio of Seeley–DeWitt coefficients fixes a universal k^4 correction to the dispersion relation,

$$\omega^2 = c^2k^2 - \gamma\ell_\chi^2 k^4 + \mathcal{O}(k^6),$$

with $\gamma = 1/(180\zeta)$ and $\zeta = \mathcal{O}(1)$ encoding scheme-dependent normalization factors. In the natural spectral scheme one finds $\gamma = 1/30$. This structure maps directly onto the parametrizations used in gravitational-wave catalogues and yields a bound on the pre-geometric scale ℓ_χ from interferometric data.

The Lorentzian completion thus embeds the spectral entropy dynamics in a fully causal framework and establishes gravitational waves as infrared probes of the underlying pre-geometric scale.

Keywords: Spectral geometry, Emergent gravity, Lorentzian geometry, Gravitational waves, Modified dispersion relations, Effective field theory, Heat-kernel expansion, Seeley–DeWitt coefficients, Causal propagation, Quantum gravity phenomenology

Contents

1 Introduction

In previous work, we introduced a spectral entropy functional $S_\Pi[g] = \frac{1}{2}\log\det{}' A_g$, where A_g is a Laplace-type elliptic operator associated with the projective substrate. In the infrared regime, the renormalized first variation of this functional was shown to generate an effective Einstein response, with higher-derivative, non-local, and anomaly contributions suppressed by powers of the pre-geometric scale ℓ_χ. This established the emergence of Einstein gravity as the dominant two-derivative sector of the spectral dynamics.

However, the construction of Papers 1 and 2 was formulated in a Riemannian (elliptic) framework. While sufficient to derive the infrared field equations and their thermodynamic interpretation, this setting does not by itself address causal propagation. In particular, the elliptic operator A_g encodes the spatial correlation structure of the substrate, but gravitational waves are dynamical, hyperbolic excitations propagating along null cones.

The purpose of the present work is therefore to provide the Lorentzian completion of the spectral entropy dynamics. We define the Lorentzian action as

$$S_\Pi^{(L)}[g] = \frac{1}{2}\Re\log\det{}'\mathcal{D}_g,$$

equivalently via a Schwinger–Keldysh prescription, so that the quadratic kernel is real and retarded. The inverse operator entering the second variation $\delta^2 S_\Pi^{(L)}$ is thus the retarded Green operator G_R, ensuring causal propagation.

We show that, in the regime $R\ell_\chi^2 \ll 1$, the quadratic operator derived from $\delta^2 S_\Pi^{(L)}$ propagates exactly two transverse-traceless tensor modes of helicity ± 2. Additional poles generated by higher-derivative corrections appear at $k^2 \sim \ell_\chi^{-2}$ and therefore lie outside the infrared window $\omega \ll \ell_\chi^{-1}$ relevant to present observations. In this domain, the theory reproduces the polarization content of general relativity.

Beyond the leading Einstein sector, the spectral expansion generates a Weyl-squared contribution whose coefficient is fixed by the Seeley–DeWitt coefficient a_4. The Lorentzian signature of the background metric is not introduced as an independent postulate: it follows from the Born–Infeld admissibility constraints of the Cosmochrony framework, as established in [1]. Linearization about Minkowski spacetime then yields a modified dispersion relation of the form

$$\omega^2 = c^2 k^2 - \gamma\,\ell_\chi^2 k^4 + \mathcal{O}(k^6),$$

where $\gamma = 1/(180\zeta)$ and $\zeta = \mathcal{O}(1)$ encodes scheme-dependent normalization factors. In the natural spectral identification of the renormalization scale with the cutoff, $\mu = \ell_\chi^{-1}$, one has $\zeta = 1/6$ and thus $\gamma = 1/30$.

This structure leads to a direct observational prediction. Gravitational-wave catalogues constrain modified dispersion relations of the form $\omega^2 = c^2k^2 + A_4 k^4$. Comparison with the spectral result yields a bound

$$\ell_\chi \lesssim \sqrt{180\,\zeta\,A_4^{obs}},$$

which becomes $\ell_\chi \lesssim \sqrt{30\,A_4^{obs}}$ in the natural scheme. The Lorentzian completion therefore provides the first direct gravitational-wave constraint on the pre-geometric scale introduced in the spectral framework.

The structure of the paper is as follows. Section 2 defines the causal Lorentzian spectral action and its second variation. Section 3 establishes the infrared propagating degrees of freedom and proves the transverse-traceless proposition. Section 4 derives the modified dispersion relation and its connection to the Seeley–DeWitt coefficients. Section 5 discusses the confrontation with gravitational-wave observations and the resulting bounds on ℓ_χ.

2 Lorentzian Completion of the Spectral Entropy Functional

2.1 From Elliptic to Hyperbolic Dynamics

In the Riemannian formulation developed previously, the spectral entropy functional is defined by

$$S_{\Pi}[g] = \frac{1}{2} \log \det{}' A_g,$$

where A_g is a Laplace-type elliptic operator. Its renormalized first variation generates an infrared Einstein response, with higher-derivative corrections governed by the Seeley–DeWitt coefficients.

While this formulation captures the static and quasi-static structure of the emergent geometry, it does not address causal propagation. In particular, A_g is elliptic and therefore encodes spatial correlation structure rather than hyperbolic time evolution. To describe gravitational waves and dynamical perturbations, a Lorentzian completion is required.

We therefore introduce a hyperbolic operator $\mathcal{D}_g$ such that its spatial restriction on constant-time slices reduces to A_g. In a $3+1$ decomposition with lapse N and shift N^i, the d'Alembertian reads

$$\Box_g = -\frac{1}{N^2}(\partial_t - \mathcal{L}_{\vec{N}})^2 + \Delta_\gamma + (\text{connection terms}),$$

where Δ_γ is the Laplacian on spatial slices. In the quasi-static regime $N \simeq 1$, $N^i \simeq 0$, and for slowly varying extrinsic curvature, one recovers

$$\Box_g \;\longrightarrow\; -\partial_t^2 + \Delta_\gamma.$$

The elliptic operator A_g of the Riemannian theory is thus identified with the spatial block of the Lorentzian operator.

2.2 Definition of the Lorentzian Spectral Action

The Lorentzian completion of the spectral entropy functional is defined as

$$S_{\Pi}^{(L)}[g] = \frac{1}{2} \Re \log \det{}' \mathcal{D}_g.$$

Equivalently, one may formulate the theory using a Schwinger–Keldysh contour prescription, so that the inverse operator entering functional variations is the retarded Green operator G_R.

This choice ensures:

- Reality of the quadratic kernel.
- Causal propagation, i.e. $G_R(x, y) = 0$ for $x \notin J^+(y)$.
- Consistency with classical gravitational-wave dynamics.

The operator $\mathcal{D}_g$ is assumed to be globally hyperbolic and normally hyperbolic in the sense of Lorentzian geometry, so that retarded and advanced Green operators exist and are unique.

2.3 Second Variation and Quadratic Operator

The second variation of the Lorentzian spectral action takes the operatorial form

$$\delta^2 S_{\Pi}^{(L)} = \frac{1}{2}\mathrm{Tr}\left(G_R\,\delta^2\mathcal{D}_g\right) - \frac{1}{2}\mathrm{Tr}\left(G_R\,\delta\mathcal{D}_g\,G_R\,\delta\mathcal{D}_g\right).$$

This expression is the hyperbolic analogue of the elliptic formula derived previously. The replacement $A_g^{-1} \to G_R$ is the essential modification ensuring causal structure.

The existence and uniqueness of retarded Green operators for normally hyperbolic operators on globally hyperbolic spacetimes are standard results of Lorentzian analysis [2]. The use of a Schwinger–Keldysh prescription ensures a causal and real effective action in curved spacetime [3, 4].

Expanding about Minkowski spacetime,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu},$$

one obtains a quadratic operator of the schematic form

$$\mathcal{O} = c_{\mathrm{EH}}\Box + \beta\Box^2 + (\text{non-local terms}) + \mathcal{O}(R\ell_\chi^2).$$

In the infrared regime $R\ell_\chi^2 \ll 1$, the dominant local contributions are the Einstein term and the Weyl-squared (Bach) correction. Non-local contributions are suppressed by additional powers of ℓ_χ^2 and are analytic in $\Box$ at low frequency.

2.4 Infrared Consistency

The structure above guarantees that:

1. The leading kinetic operator is hyperbolic and luminal.
2. Higher-derivative corrections introduce an additional pole at $k^2 \sim \ell_\chi^{-2}$.
3. In the domain $\omega \ll \ell_\chi^{-1}$, the additional excitation decouples, leaving a standard massless graviton sector.

The Lorentzian completion therefore preserves the infrared Einstein limit while embedding it in a fully causal framework. Subsequent sections analyze the propagating degrees of freedom and derive the resulting modified dispersion relation.

3 Infrared Spectrum and Propagating Degrees of Freedom

3.1 Linearization Around Minkowski Spacetime

We consider perturbations about flat spacetime,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu},$$

and work to quadratic order in $h_{\mu\nu}$. In the infrared regime $R\ell_\chi^2 \ll 1$, the quadratic operator derived from $\delta^2 S_{\Pi}^{(L)}$ takes the local form

$$\mathcal{O}_{\mu\nu}{}^{\rho\sigma} = c_{\mathrm{EH}}\Box + \beta\Box^2 + \mathcal{O}(\ell_\chi^4\Box^3),$$

where $c_{\mathrm{EH}} > 0$ and $\beta < 0$ in the natural spectral scheme.

We impose harmonic gauge,

$$\partial^\mu \bar{h}_{\mu\nu} = 0, \qquad \bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h,$$

which eliminates gauge redundancies at linear order.

3.2 Scalar–Vector–Tensor Decomposition

To identify the propagating modes, we perform a scalar–vector–tensor (SVT) decomposition with respect to spatial rotations. Writing spatial indices as $i, j = 1, 2, 3$, the perturbation splits as

$$h_{00}, \qquad h_{0i} = \partial_i B + B_i, \qquad h_{ij} = 2\psi\,\delta_{ij} + 2\partial_i\partial_j E + \partial_i E_j + \partial_j E_i + h_{ij}^{\mathrm{TT}},$$

with

$$\partial_i B_i = 0, \qquad \partial_i E_i = 0, \qquad \partial_i h_{ij}^{\mathrm{TT}} = 0, \qquad h_{ii}^{\mathrm{TT}} = 0.$$

In harmonic gauge, the scalar and vector components enter the quadratic action without independent kinetic terms in the regime $R\ell_\chi^2 \ll 1$. They are therefore non-dynamical and can be eliminated algebraically.

The only sector acquiring a propagating kinetic operator is the transverse-traceless tensor sector h_{ij}^{TT}.

3.3 Transverse-Traceless Sector

Restricting to $h_{\mu\nu}^{\mathrm{TT}}$, the quadratic operator reduces to

$$\left(c_{\mathrm{EH}}\Box + \beta\Box^2\right) h_{\mu\nu}^{\mathrm{TT}} = 0.$$

Factoring gives

$$\Box\left(1 + \frac{\beta}{c_{\mathrm{EH}}}\Box\right) h_{\mu\nu}^{\mathrm{TT}} = 0.$$

The first factor corresponds to the standard massless graviton pole $k^2 = 0$. The second factor introduces an additional pole at

$$k^2 = -\frac{c_{\mathrm{EH}}}{\beta} \sim \ell_\chi^{-2}.$$

Since ℓ_χ^{-1} sets the pre-geometric ultraviolet scale, this excitation lies outside the infrared window

$$\omega \ll \ell_\chi^{-1},$$

relevant to gravitational-wave observations.

3.4 Infrared Propagation Theorem

Proposition. In the regime $R\ell_\chi^2 \ll 1$ and $\omega \ll \ell_\chi^{-1}$, the Lorentzian spectral action propagates exactly two transverse-traceless tensor modes of helicity ± 2. No scalar or vector propagating degree of freedom is dynamically accessible in this domain.

Proof. The SVT decomposition shows that scalar and vector components lack independent kinetic operators in harmonic gauge. The tensor sector obeys a fourth-order equation that factorizes into two second-order operators. The additional pole is located at $k^2 \sim \ell_\chi^{-2}$, hence outside the infrared regime. Only the massless pole contributes to low-frequency propagation. □

3.5 Polarization Content

The surviving infrared excitations are precisely the two helicity ± 2 tensor modes of general relativity. No scalar (breathing or longitudinal) or vector polarizations appear in the accessible frequency range. The Lorentzian completion of the spectral framework is therefore consistent with current gravitational-wave polarization constraints.

4 Modified Dispersion from Spectral Coefficients

4.1 Quadratic Operator and Pole Structure

In the transverse-traceless sector established in Section 3, the quadratic operator takes the factorized form

$$\mathcal{O}_{\mathrm{TT}} = c_{\mathrm{EH}}\Box + \beta\Box^2 = c_{\mathrm{EH}}\Box\left(1 + \frac{\beta}{c_{\mathrm{EH}}}\Box\right).$$

The coefficient β originates from the Weyl-squared term in the renormalized spectral expansion. For a Laplace-type scalar minimal operator in four dimensions, the Seeley–DeWitt coefficient a_4 contains

$$\alpha_C = \frac{1}{360(4\pi)^2},$$

multiplying $C^2_{\mu\nu\rho\sigma}$. Using

$$\delta\int\sqrt{g}\,C^2 = -2\int\sqrt{g}\,B_{\mu\nu}\,\delta g^{\mu\nu},$$

one obtains

$$\beta = -2\alpha_C = -\frac{1}{180(4\pi)^2}.$$

The heat-kernel expansion and the associated Seeley–DeWitt coefficients are reviewed in [6, 5]. The variation of the Weyl-squared term leading to the Bach tensor is standard in higher-derivative gravity [7, 8].

The Einstein prefactor is parameterized as

$$c_{\mathrm{EH}} = \zeta\,(4\pi)^{-2}\,\ell_\chi^{-2},$$

where $\zeta = \mathcal{O}(1)$ accounts for scheme-dependent normalizations and possible multiplicity factors. In the natural spectral identification of the renormalization scale with the cutoff, $\mu = \ell_\chi^{-1}$, and using $a_2 = R/6$, one has $c_{\mathrm{EH}} = \mu^2/[6(4\pi)^2]$, corresponding to $\zeta = 1/6$.

4.2 Infrared Dispersion Relation

For plane-wave solutions $h^{\mathrm{TT}}_{\mu\nu} \sim e^{ik\cdot x}$, the operator yields the dispersion equation

$$-c_{\mathrm{EH}}k^2 + \beta k^4 = 0.$$

Solving for ω^2 gives

$$\omega^2 = c^2k^2 - \frac{\beta}{c_{\mathrm{EH}}}k^4.$$

Using the expressions above,

$$\frac{\beta}{c_{\rm EH}} = -\frac{1}{180\,\zeta}\,\ell_\chi^2,$$

so that the infrared dispersion relation becomes

$$\omega^2 = c^2k^2 - \gamma\ell_\chi^2 k^4 + \mathcal{O}(k^6),$$

with

$$\gamma = \frac{1}{180\,\zeta}.$$

In the natural spectral scheme $\zeta = 1/6$,

$$\gamma = \frac{1}{30},$$

and therefore

$$\omega^2 = c^2k^2 - \frac{1}{30}\,\ell_\chi^2 k^4 + \mathcal{O}(k^6).$$

4.3 Physical Interpretation

The correction term is negative, implying a slight sub-luminal deviation of the group velocity at high frequency,

$$v_g = \frac{\partial\omega}{\partial k} \simeq c\left(1 - \frac{3}{2}\gamma\ell_\chi^2 k^2\right).$$

Since the modification is suppressed by $(k\ell_\chi)^2$, it is negligible in the regime $k\ell_\chi \ll 1$. The additional pole at $k^2 \sim \ell_\chi^{-2}$ corresponds to a heavy excitation beyond the infrared window probed by current interferometers.

The modified dispersion relation is therefore fully consistent with the infrared Einstein limit, while providing a concrete k^4 correction that can be directly confronted with gravitational-wave observations.

5 Confrontation with Gravitational-Wave Observations

5.1 Mapping to the LVK Parametrization

Gravitational-wave catalogues constrain modified dispersion relations of the general form

$$\omega^2 = c^2k^2 + A_\alpha k^\alpha,$$

with $\alpha = 4$ corresponding to the leading higher-derivative correction compatible with locality and parity.

The spectral Lorentzian completion derived above predicts

$$\omega^2 = c^2k^2 - \gamma\ell_\chi^2 k^4 + \mathcal{O}(k^6),$$

with

$$\gamma = \frac{1}{180\,\zeta}.$$

Identifying coefficients gives

$$A_4 = -\gamma\ell_\chi^2.$$

Current analyses place bounds on $|A_4|$ from phase corrections accumulated during propagation over cosmological distances. Denoting the observational upper bound by A_4^{obs}, the spectral prediction implies

$$\ell_\chi \lesssim \sqrt{\frac{A_4^{obs}}{\gamma}} = \sqrt{180\,\zeta\,A_4^{obs}}.$$

In the natural spectral scheme $\zeta = 1/6$, this reduces to

$$\ell_\chi \lesssim \sqrt{30\,A_4^{obs}}.$$

5.2 Robustness and Scheme Dependence

The coefficient $\zeta = \mathcal{O}(1)$ captures normalization conventions in the identification of the Einstein term and possible multiplicity factors associated with the underlying operator. While its precise numerical value is scheme-dependent, the following features are robust:

- The sign of the k^4 correction is negative.
- The structure of the dispersion is fixed by the ratio of Seeley–DeWitt coefficients.
- The constraint on ℓ_χ scales as $\sqrt{\zeta}$ and is therefore stable up to factors of order unity.

Thus, gravitational-wave observations provide a direct infrared probe of the pre-geometric scale introduced in the spectral framework.

5.3 Luminal Front Propagation

Multi-messenger observations constrain the difference between the speed of gravitational waves and the speed of light to extremely high precision. In the present framework, the characteristic operator remains $\Box$ at leading order, and the correction enters only at order k^4. The front velocity is therefore luminal, and deviations of the group velocity are suppressed by $(k\ell_\chi)^2$.

Consequently, the Lorentzian spectral completion is consistent with current bounds on gravitational-wave speed while predicting a specific higher-order dispersive correction.

5.4 Summary of Observational Implications

Modified dispersion relations for gravitational waves are constrained by the analyses of the [**?**] and subsequent catalogues of the [**?**, **?**]. Multi-messenger observations further constrain deviations from luminal propagation [**?**].

The Lorentzian completion of the spectral entropy functional leads to:

1. Exactly two propagating tensor polarizations in the infrared.
2. A modified dispersion relation with fixed structure $\omega^2 = c^2k^2 - \gamma\ell_\chi^2 k^4$.
3. A direct observational bound on the pre-geometric scale ℓ_χ from gravitational-wave data.

This establishes, for the first time within the spectral framework, a concrete and testable connection between the emergent geometric dynamics and interferometric observations.

6 Conclusion

We have constructed the Lorentzian completion of the spectral entropy framework introduced in previous work. While the original formulation was elliptic and sufficient to derive the infrared Einstein response and its thermodynamic interpretation, it did not address causal propagation. The present paper establishes that the spectral dynamics admits a consistent hyperbolic completion with a well-defined retarded Green operator and a real quadratic kernel.

Linearization about Minkowski spacetime shows that, in the regime $R\ell_\chi^2 \ll 1$ and $\omega \ll \ell_\chi^{-1}$, the theory propagates exactly two transverse-traceless tensor modes of helicity ± 2. Scalar and vector sectors remain non-dynamical in the infrared, and the additional pole generated by higher-derivative corrections appears at $k^2 \sim \ell_\chi^{-2}$, beyond the frequency range of current gravitational-wave observations. The polarization content therefore coincides with that of general relativity in the accessible domain.

Beyond the leading Einstein sector, the ratio of Seeley–DeWitt coefficients fixes a universal k^4 correction to the dispersion relation,

$$\omega^2 = c^2k^2 - \gamma\ell_\chi^2 k^4 + \mathcal{O}(k^6),$$

with $\gamma = 1/(180\zeta)$ and $\zeta = \mathcal{O}(1)$ encoding scheme-dependent normalization factors. In the natural spectral scheme, $\gamma = 1/30$. This structure provides a direct mapping to the parametrizations used in gravitational-wave catalogues and yields a bound on the pre-geometric scale ℓ_χ from observational data.

The Lorentzian completion therefore closes the logical chain of the spectral program: the elliptic structure encodes spatial correlations, the renormalized first variation yields the infrared Einstein response, and the hyperbolic extension ensures causal propagation and observational testability. Gravitational waves thus provide a concrete infrared probe of the pre-geometric scale underlying the spectral construction.

Further developments include a fully covariant coupling to matter sectors and a systematic analysis of non-local corrections beyond the leading k^4 term. These directions may sharpen the connection between the spectral substrate and precision gravitational observations.

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